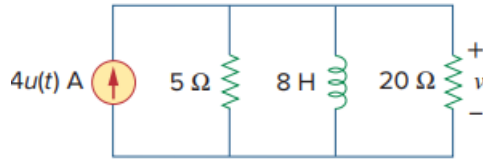


Problem

Find $v(t)$ for $t > 0$ in the circuit of Fig. 7.125 if the initial current in the inductor is zero.

Fig. 7.125:



Step-by-step solution

Step 1 of 8

Refer to Figure 7.125 in the textbook.

The current source is, $i_s = 4u(t)$ A.

The input current source is defined as,

$$i_s = \begin{cases} 0 \text{ A} & \text{for } t < 0 \\ 4 \text{ A} & \text{for } t > 0 \end{cases}$$

The initial current in the inductor is zero.

That is,

$$i_L(0) = 0$$

Step 2 of 8

Draw the circuit diagram at $t = 0^+$ and replace the inductor with initial current.

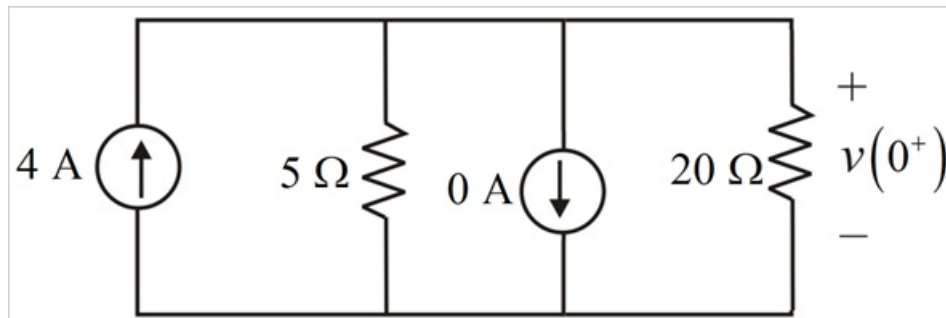


Figure 1

Step 3 of 8

Calculate the current through $20\ \Omega$ resistor.

$$i_{20\ \Omega} = 4 \left(\frac{5}{20+5} \right) \\ = 0.8\ \text{A}$$

Calculate the initial voltage across the $20\ \Omega$ resistor.

$$v(0^+) = 20(0.8) \\ = 16\ \text{V}$$

Step 4 of 8

Draw the circuit diagram for $t > 0$.

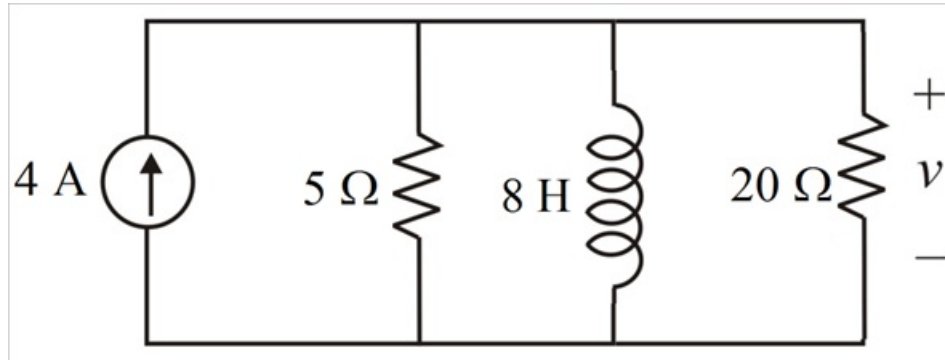


Figure 2

Step 5 of 8

Calculate the time constant.

$$\tau = \frac{L}{R_{eq}}$$

Here,

R_{eq} is equivalent resistance seen across inductor terminal by open circuiting the current source.

Draw the circuit diagram to calculate the equivalent resistance.

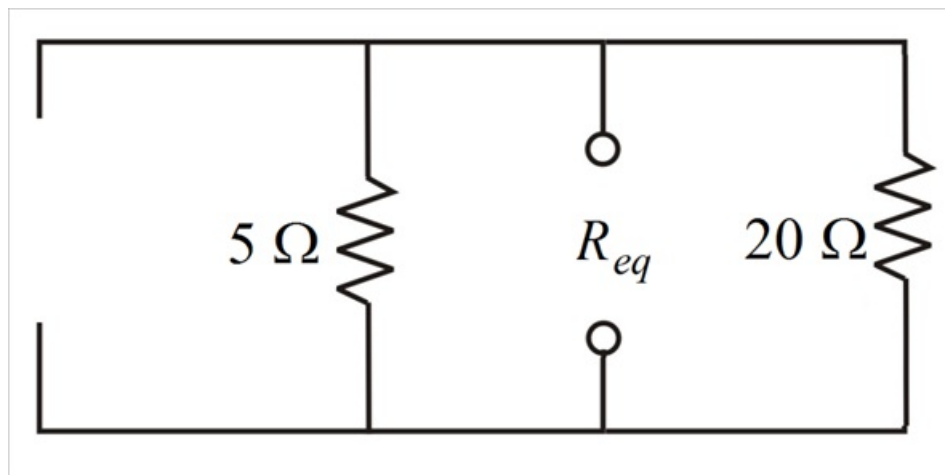


Figure 3

Step 6 of 8

From Figure 3,

$$\begin{aligned} R_{eq} &= (5) \parallel (20) \\ &= \frac{(5)(20)}{5+20} \\ &= 4 \, \Omega \end{aligned}$$

Therefore, the time constant is,

$$\begin{aligned} \tau &= \frac{8}{4} \\ &= 2 \text{ sec} \end{aligned}$$

Step 7 of 8

Draw the circuit diagram at $t = \infty$.

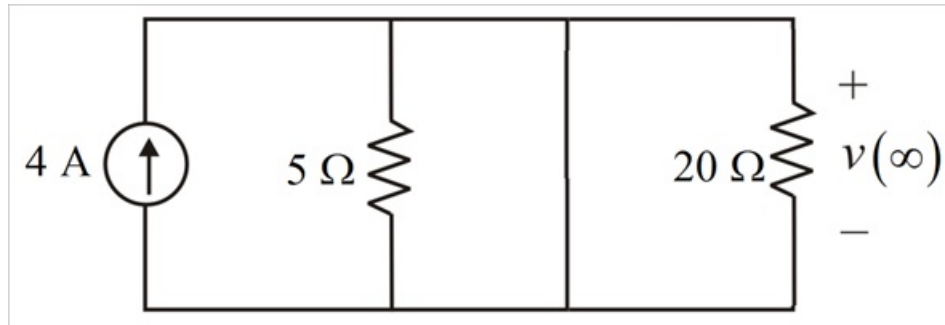


Figure 4

Step 8 of 8

In Figure 4, the total current is flowing through the short circuit only. Therefore, the voltage is,

$$v(\infty) = 0 \text{ V}$$

The voltage across the resistor is,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{\frac{-t}{\tau}}$$

Substitute 16 V for $v(0)$, 0 V for $v(\infty)$ and 2 s for τ .

$$\begin{aligned} v(t) &= 0 + [16 - 0]e^{\frac{-t}{2}} \\ &= 16e^{-0.5t} \text{ V} \end{aligned}$$

Therefore, the voltage across the resistor, $v(t)$ is $16e^{-0.5t} \text{ V}$.